

13 B

Isothermal Process: Since $T_B = T_C$,

use $PV = \text{constant}$

$$200 \times V_B = 500 \times 2, V_B = 5 \text{ m}^3.$$

Net work done is area enclosed by ABC in the PV graph.

Net WD = + ve

$$\approx (0.5 \times (500-200) \times 3) \text{ (approximate)}$$

Net work done on gas is less than +450 kJ